

Refer "laptops.csv" and perform following data wrangling operations

1. Evaluate the dataset for missing data and replace with mean
2. Convert "Weight_kg" and "Screen_Size_cm" into appropriate format
3. Standardize the value of Screen_size to inches and weight of the laptop in pounds.
4. Write a code to create 3 bins for the attribute "Price". These bins would be named "Low", "Medium" and "High". The new attribute will be named "Price-binned".
5. Turn categorical variable "Screen" into quantitative variables.

```
# Import necessary libraries
```

```
import pandas as pd
```

```
import numpy as np
```

```
# Load the dataset
```

```
df = pd.read_csv("laptops.csv")
```

```
# Step 1: Evaluate dataset for missing data and replace with mean
```

```
print("Missing values in each column:")
```

```
print(df.isnull().sum())
```

```
# Replace missing values with the mean of each column (for numeric columns)
```

```
df.fillna(df.mean(), inplace=True)
```

```
# Step 2: Convert "Weight_kg" to pounds and "Screen_Size_cm" to inches
```

```
# 1 kg = 2.20462 lbs
```

```
# 1 cm = 0.393701 inches
```

```
df['Weight_lb'] = df['Weight_kg'] * 2.20462
```

```
df['Screen_Size_inch'] = df['Screen_Size_cm'] * 0.393701
```

```
# Step 3: Standardize values

# Already converted to the required units: pounds and inches

# So we skip the actual transformation here as it's already done in step 2.


# Step 4: Create 3 bins for "Price" into "Low", "Medium", and "High"
price_bins = [0, 500, 1500, np.inf] # Define bin edges
price_labels = ['Low', 'Medium', 'High'] # Define bin labels
df['Price-binned'] = pd.cut(df['Price'], bins=price_bins, labels=price_labels, right=False)


# Step 5: Convert the "Screen" categorical variable into quantitative (one-hot encoding)
df = pd.get_dummies(df, columns=['Screen'], drop_first=True) # One-hot encoding


# Display the transformed dataset
print("\nTransformed Dataset:")
print(df.head())
```